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Pandas Cheat Sheet

Essential DataFrame operations for data science & analysis.

CREATING DATAFRAMES

```
pd.DataFrame({'a':[1,2],'
pd.read_csv('file.csv')
pd.read_excel('file.xlsx'
pd.read_json('file.json')
pd.read_sql(query, conn)
pd.DataFrame(np.random.ra
```

INSPECTION

```
df.head(n)
df.tail(n)
df.shape
df.info()
df.describe() # stats sum
df.dtypes
df.columns
```

SELECTION

```
df['col']
df[['a','b']]
df.loc[0]
df.iloc[0]
df.loc[0,'col']
df.iloc[0:3,1:3] # slice
df.at[0,'col']
```

FILTERING

```
df[df['a'] > 5]
df[df['a'].isin([1,2,3])]
df[df['a'].between(1,5)]
df[(df.a>1) & (df.b<5)]
df[(df.a>1) | (df.b<5)]
df[~(df['a'] > 5)]
df.query('a > 5 & b < 3')
```

MISSING VALUES

```
df.isnull().sum()
df.dropna()
df.dropna(axis=1)
df.fillna(0)
df.fillna(method='ffill')
df.fillna(df.mean())
df.interpolate()
```

SORTING & RANKING

```
df.sort_values('col')
df.sort_values('col',asce
df.sort_values(['a','b'])
df.sort_index()
df['col'].rank()
df.nlargest(5,'col')
df.nsmallest(5,'col')
```

GROUPBY & AGGREGATION

```
df.groupby('col').mean()
df.groupby('col').agg('su
df.groupby('col').agg(['m
df.groupby(['a','b']).siz
df.groupby('col').apply(f
df.pivot_table(values='v'
  index='r', columns='c')
```

MERGING & JOINING

```
pd.merge(df1,df2,on='key'
pd.merge(df1,df2,how='lef
pd.merge(df1,df2,how='out
pd.concat([df1,df2])
pd.concat([df1,df2],axis=
df1.join(df2,on='key')
```

TRANSFORMING

```
df['col'].apply(fn)
df.apply(fn, axis=1)
df.applymap(fn)
df.map({'a':1,'b':2})
df.assign(new=df.a+df.b)
df.rename(columns={'old':
df.drop(columns=['col'])
```

STRING OPERATIONS

```
df['col'].str.lower()
df['col'].str.upper()
df['col'].str.strip()
df['col'].str.replace('a'
df['col'].str.contains('p
df['col'].str.split(',',e
df['col'].str.len()
```

RESHAPING

```
df.melt(id_vars=['id'])
df.pivot(index='r',column
df.stack()
df.unstack()
df.T
df.explode('col')
df.reset_index(drop=True)
```

EXPORTING

```
df.to_csv('out.csv',index
df.to_excel('out.xlsx')
df.to_json('out.json')
df.to_parquet('out.parque
df.to_sql('table', conn)
df.to_clipboard()
df.to_dict()
```